

No. of Functional Features Included in the Solution

Date	3 July 2026
Team ID	XXXXXX
Project Name	OptiCrop – Smart Agricultural Production Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status (Done / In Progress / Pending)	Marks Contribution
1	User Input Form	Allows users to enter soil nutrients (N, P, K), temperature, humidity, pH, and rainfall values.	Frontend (HTML/CSS)	Done	High
2	Crop Prediction	Predicts the most suitable crop using the trained Machine Learning model.	ML Model	Done	High
3	Flask Backend	Processes user input, validates data, and communicates with the ML model.	Backend (Flask)	Done	High
4	Prediction Result Display	Displays the recommended crop instantly on the web page	Frontend + Backend	Done	Medium
5	Input Validation	Validates user input to avoid invalid or missing values.	Backend	Done	Medium
6	Responsive User Interface	Provides a clean and user-friendly interface accessible from different screen sizes.	Frontend	Done	Medium
7	Model Integration	Loads the trained model.pkl file and generates predictions.	ML Integration	Done	High
8	Future Enhancement Support	Architecture supports future integration of weather API, fertilizer recommendation, and disease detection..	System Design	In Progress	Low

Feature Summary:

Metric	Count / Value
Total Features Planned	8
Total Features Implemented	7
Core / Must-Have Features	5

Additional / Nice-to-Have Features	3
Features Tested & Verified	7

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Input Form, Responsive Design, Result Display
2	Backend / Logic	3	Flask Backend, Input Validation, Prediction Processing
3	Database / Storage	2	Crop Dataset (CSV), Trained ML Model (model.pkl)
4	API / Integration	1	Machine Learning Model Integration
5	Security / Authentication	1	Basic Input Validation (No user authentication in current version)